



Third Year Engineering

21BTCS503 - Introduction to Data Science

Class - T.Y. Core (SEM-V)

Unit - I





Unit - I: Syllabus

- Introduction to Data Science.
- Facets of Data : Structured data, Unstructured data , Natural language, Machine generated data, Graph based or network data, Audio, Image and Video, Streaming data
- The Data Science Process: setting the research goal, retrieving data, Data preparation, Data exploration, model building, Presentation and Automation.





What is data?

- Data is nothing but the collection of information, facts, or statistics that you can study and understand. You can use data across various fields and industries to guide decision-making, improve processes, and find valuable insights.

Facets of Data:

- Structured data, Unstructured data , Natural language, Machine generated data, Graph based or network data, Audio, Image and Video, Streaming data



Facets of Data:

Structured data: Structured data is having a predefined format and it is arranged in rows and column format. Structured data is identifiable because it is organized in a structure. It helps for application to retrieve and process data easily. Database management system is used for storing structured data.

E.g. Spreadsheet data: , Relational data, Transactional data, Sensor data, Excel table

Unstructured data: Unstructured data is data that does not have a predefined structure or specified format and it does not fit well into traditional databases. Row and columns are not used for unstructured data. Unstructured data has no identifiable structure. Therefore it is difficult to retrieve required information.,

E.g. Textual data, Documents, email messages, Multimedia data, web data, social media dat.



Natural language: Natural language is a special type of unstructured data. Natural language processing enables machines to recognize characters, words and sentences, then apply meaning and understanding to that information. This helps machines to understand language as humans do.

For natural language processing to help machines understand human language, it must go through speech recognition, natural language understanding and machine translation. It is an iterative process comprised of several layers of text analysis.





Machine generated data: Machine-generated data is an information that is created without human interaction as a result of a computer process or application activity. This means that data entered manually by an end-user is not recognized to be machine-generated. Machine data contains a definitive record of all activity and behavior of our customers, users, transactions, applications, servers, networks, factory machinery and so on. Both Machine-to-Machine (M2M) and Human-to-Machine (H2M) interactions generate machine data. Machine data is generated continuously by every processor-based system, as well as many consumer-oriented systems. Examples of machine data are web server logs, call detail records, network event logs and telemetry.



Graph based or network data :

Graph-based or network-based data refers to data that represents relationships or connections between entities. In this type of data, entities are represented as nodes or vertices, and the relationships between them are represented as edges or links. Graph databases are used to store graph-based data and are queried with specialized query languages such as SPARQL.

E.g. Recommendation systems, Transportation systems





Audio, Image and Video : The terms audio and video commonly refers to the time-based media storage format for sound/music and moving pictures information. It is important to remark that multimedia data is one of the most important sources of information and knowledge

Streaming data: Streaming data is data that is generated continuously by thousands of data sources, which typically send in the data records simultaneously and in small sizes (order of Kilobytes)





Types Of Data

The two main types of data are: **Qualitative Data** and **Quantitative Data**

The **Qualitative data** are further classified into two parts:

- **Nominal data**
- **Ordinal data**

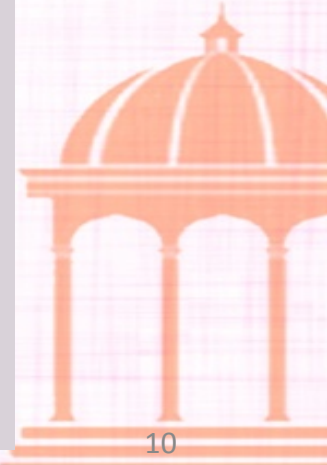
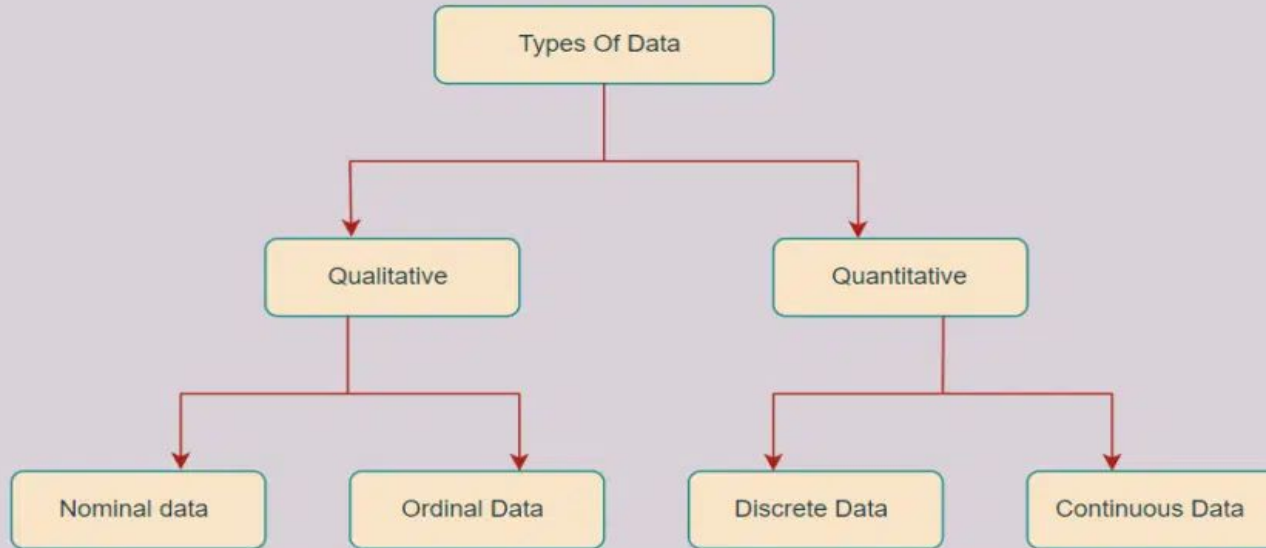
The **Quantitative data** are further classified into two parts:

- **Discrete data**
- **Continuous data**





Types of Data





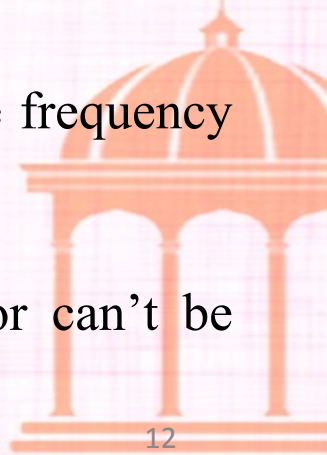
Qualitative Data

- Qualitative data is a type of data that can't be measured or counted in the form of numbers.
- It is non-numerical information describing qualities, characteristics, or opinions.
- It is also known as the categorical data. These types of data are sorted by category, not by number. That's why it is also known as Categorical Data.
- The categorical information involves categorical variables that describe the features such as a person's gender, home town, favourite sport etc.
- Sometimes categorical data can hold numerical values (quantitative value), but those values do not have a mathematical sense. Examples birthdate, school postcode.
- This data helps market researchers understand the customers' tastes and then design their ideas and strategies accordingly.
- The Qualitative data are further classified into two parts Nominal data and Ordinal data



Nominal Data:

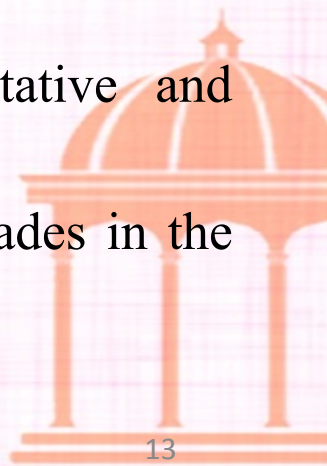
- Nominal data is one of the types of qualitative information which helps to label the variables without any order or numerical value or quantitative value.
- It cannot be ordered and measured
- The name “nominal” comes from the Latin name “nomen,” which means “name.”
- In this method, the data are grouped into categories, and then the frequency or the percentage of the data can be calculated.
- Examples include your gender, nationality, or hair color
- The color of hair can be considered nominal data, as one color can't be compared with another color.





Ordinal Data:

- Ordinal data represents information that follows a clear order or ranking, where a number is present in some kind of order by their position on the scale.
- But the differences between the data values are not quantifiable and we can't do any arithmetical tasks on them.
- These kinds of data can be considered “in-between” qualitative and quantitative data.
- This variable is mostly found in surveys, educational levels , grades in the exam, finance, economics, questionnaires, and so on.





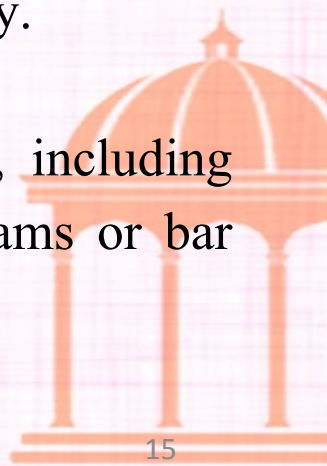
Quantitative Data

- Quantitative data is a type of data that can be expressed in numerical values, that you can measure, count, or express using numbers. These kinds of data are also known as Numerical data.
- Numerical data gives information about the quantities of a specific thing.
- This type of data allows you to perform various calculations, such as averages, correlations, and regressions, to identify patterns, trends, and relationships
- It answers the questions like “how much,” “how many,” and “how often.”
- Some examples of numerical data are temperature, price, height, length, size, weight, and so on.
- The quantitative data can be classified into two different types based on the data sets i.e. Discrete data and Continuous data.



Discrete Data:

- The term discrete means distinct or separate.
- The discrete data are countable and have finite values, their subdivision is not possible
- It consists of distinct, separate values or categories that you can measure.
- Example: Number of students in the class, employees in a company.
- These data can't be broken into decimal or fraction values.
- When analyzing discrete data, you can use summary statistics, including mean, median, and mode, as well as visualizations like histograms or bar charts to display the data distribution.





Continuous Data:

- Continuous data is data that can be calculated. It has an infinite number of probable values that can be selected within a given specific range.
- Continuous data are in the form of fractional numbers
- Continuous data represents information that can be divided into smaller levels. The continuous variable can take any value within a range.
- The key difference between discrete and continuous data is that discrete data contains the integer or whole number. Still, continuous data stores the fractional numbers to record different types of data such as temperature, height, width, time, speed, etc
- Example: Temperature range, version of an android phone, the height of a person, the length of an object

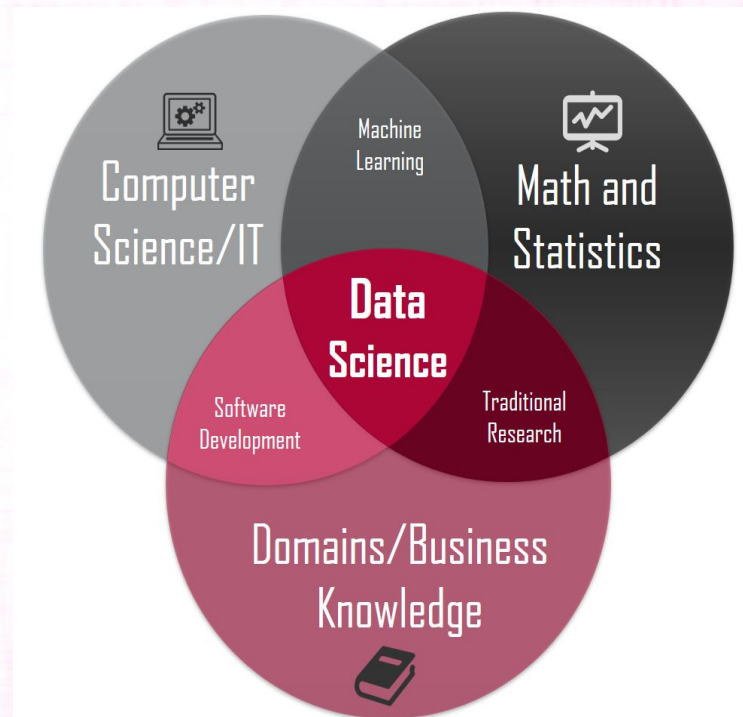
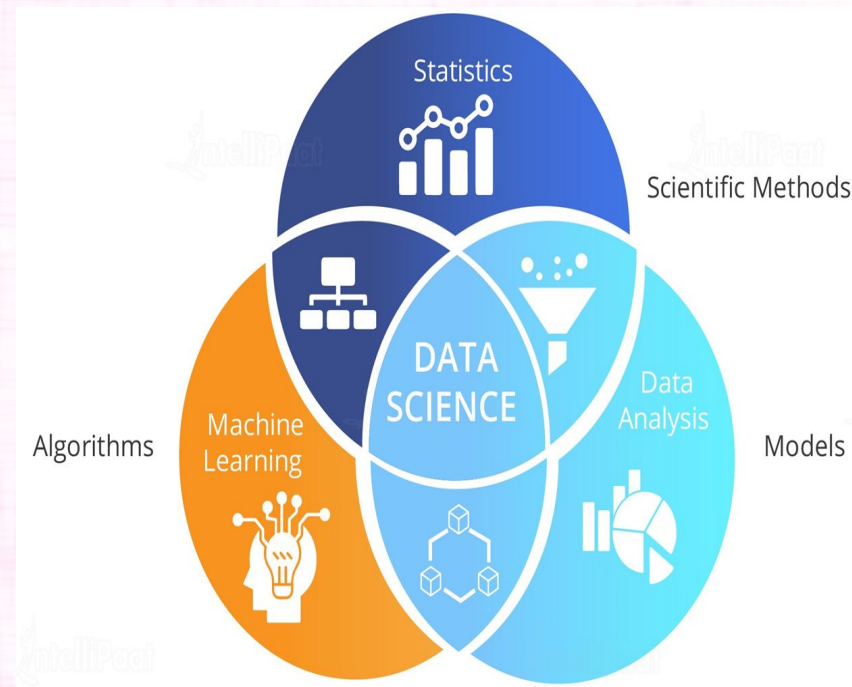


Data science is an interdisciplinary field that uses algorithms, procedures, and processes to examine large amounts of data in order to uncover hidden patterns, generate insights, and direct decision making.

Data science is a deep study of the huge amount of data, which involves extracting meaningful insights from raw, structured, and unstructured data that is processed using the scientific method, different technologies, and algorithms. It is a multidisciplinary field that uses tools and techniques to manipulate the data so that you can find something new and meaningful.



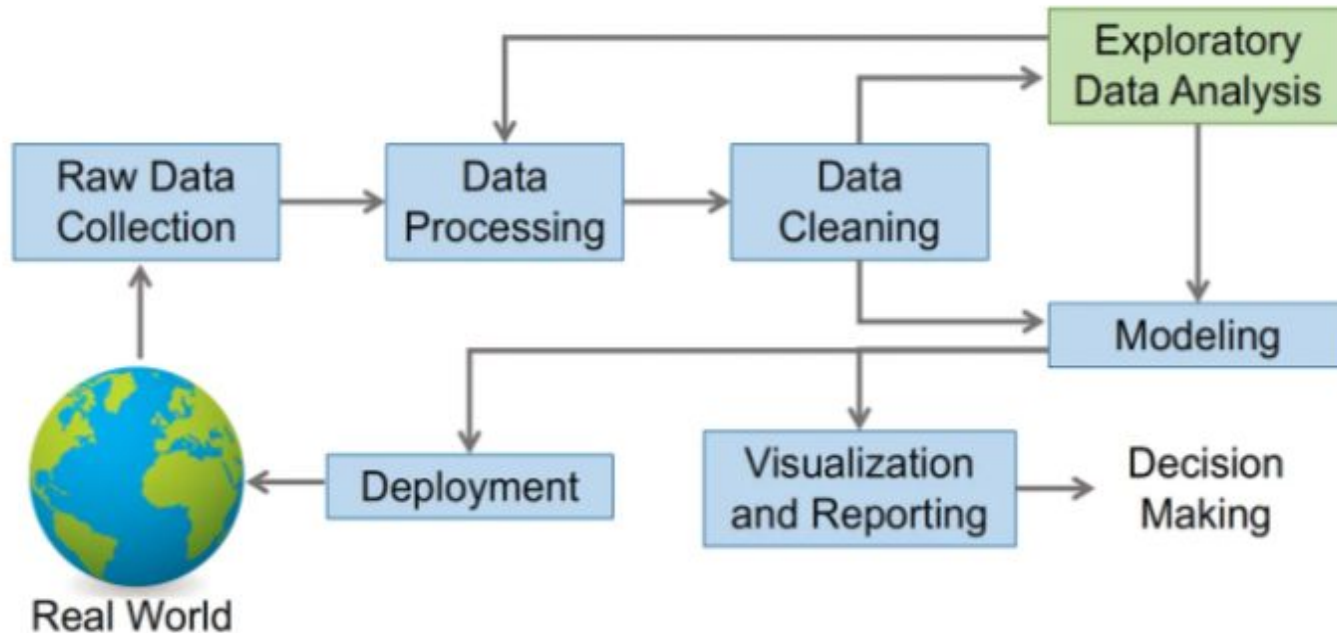
Data Science





Data Science Process.

Data Science Process





Data Science Process.

- Business understanding
- Data collection
- Data preparation
- Modeling
- Evaluation
- Deployment





Business understanding

- The first step of the process is to clarify the business's goals and bring focus to the data science project.
- Understand the business problem, its impact, the ultimate goals for addressing it, and the relevant project plan.
- To better understand the business, data scientist meet with stakeholders, subject matter experts, and others who can offer insights into the problem at hand.
- They may also do preliminary research to see how others have tried to solve similar problems. Ultimately, they'll have a clearly defined problem and a roadmap to solving it



Data Collection

- The next step is understanding your data so gather data from various sources, such as databases, APIs, or web scraping
- In this phase, you'll determine what data you have, where you can get more of it, what your data includes, and its quality.
- You'll also decide what data collection tools you'll use and how you'll collect your initial data. Then you'll describe the properties of your initial data, such as the format, the quantity, and the records or fields of your data sets.
- Collecting and describing your data will allow you to begin exploring it. You can then formulate your first hypothesis by asking data science questions that can be answered through queries, visualization, or reporting. Finally, you'll verify the quality of your data by determining if there are errors or missing values.



Data preparation

- Data preparation is often the most time-consuming phase, and you may need to revisit this phase multiple times throughout your project.
- Data comes from various sources and is usually unusable in its raw state, as it often has corrupt and missing attributes, conflicting values, and outliers. Data preparation resolves these issues and improves the quality of your data, allowing it to be used effectively in the modeling stage.
- Data preparation involves many activities that can be performed in different ways. Data cleaning, Data integration, Data transformation, Data reduction, Data discretization, Feature engineering.



Modeling

- Use the cleaned and understood data to build and train machine learning models
- There are multiple options for data modeling. You'll choose which option is best based on the goals of the business, the variables involved, and the tools available.
- Once you've chosen your modeling technique, you'll design tests to determine how well your model works. For this step, your deliverable will be your test design. This may entail splitting your data into training data and testing data to avoid over fitting, which happens when you design a model that perfectly fits one set of data but doesn't work with others.
- Next, you'll build your model to address your specific business goals. When you do this, you'll deliver three items: A list of parameter settings, A description of the models, The models themselves
- The final step in the modeling phase is to assess your models.



Evaluation

- In the evaluation phase, you'll evaluate or assess the performance of the model using appropriate evaluation metrics, such as accuracy, precision and recall based on the goals of your business.
- Review your work process and explain how your model will help the business, summarize your findings, and make any corrections.
- Finally, you'll determine your next steps. Is your model ready for deployment? Does it need a new iteration or another dependency project?



Deployment

- While deployment is the last phase, it's not necessarily the end of your project. During the deployment phase, you'll plan and document how you intend to deploy the model and how the results will be delivered and presented.
- You'll also need to monitor the results and maintain the model during the deployment phase.
- You'll conclude your project by creating a summary report and presentation. Afterward, plan to review the entire process to determine what worked and what needs improvement.



Data Science Applications/Projects deployed

- Nissans Robotic Chairs
- AI finding your dream job
- ChatGPT
- AI Powered shoes
- Ameca
- TayTweets (Why was it stopped?)
- Smart Watches
- Netflix





Data Science Applications





4 PILLARS OF DATA SCIENCE



Domain Knowledge

- > Business knowledge
- > Expert systems
- > User testing



Math & Statistics skills

- > Linear algebra
- > Calculus
- > Descriptive statistics
- > Inferential statistics



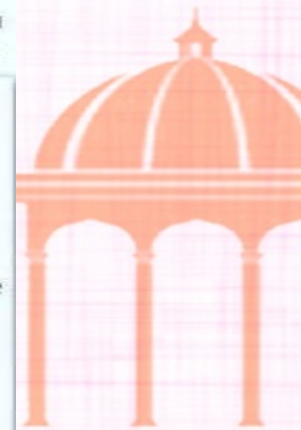
Computer science

- > Big data technologies
- > Programming
- > Database



Communication & Visualization

- > Storytelling skills
- > Visual art design
- > Able to engage with senior manager
- > R packages





Career opportunities in Data Science field

- "The rise of Data Science needs will create roughly 11.5 million job openings by 2025" US Bureau of Labour Statistics.
- "By 2026, Data Scientists and Analysts will become the number one emerging role in the world." World Economic Forum 2.
- Data Science and Artificial Intelligence are amongst the hottest fields of the 21st century that will impact all segments of daily life by 2025, from transport and logistics to healthcare and customer service.



Structuring features according to their type

- **Quantitative features** possess “numerical quantity”, such as height, age, number of births, etc., and can either be continuous or discrete.
- **Qualitative features** do not have a numerical meaning, but their possible values can be divided into a fixed number of categories, such as {M,F} for gender or {blue, black, brown, green} for eye color. For this reason such features are also called categorical. (Nominal or ordinal)



MIT School of Computing

Department of Computer Science & Engineering



MIT-ADT
UNIVERSITY
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Feature	Description	Unit or Coding
gender	Gender	1=Male; 2=Female
situation	Family status	1=Single 2=Living with spouse 3=Living with family 4=Living with someone else
tea	Daily consumption of tea	Number of cups
coffee	Daily consumption of coffee	Number of cups
height	Height	cm
weight	Weight (actually: mass)	kg
age	Age at date of interview	Years
meat	Consumption of meat	0=Never 1=Less than once a week 2=Once a week 3=2–3 times a week 4=4–6 times a week 5=Every day
fish	Consumption of fish	As in meat
raw_fruit	Consumption of raw fruits	As in meat
cooked_fruit_veg	Consumption of cooked fruits and vegetables	As in meat
chocol	Consumption of chocolate	As in meat
fat	Type of fat used for cooking	1=Butter 2=Margarine 3=Peanut oil 4=Sunflower oil 5=Olive oil 6=Mix of vegetable oils (e.g., Isio4) 7=Colza oil 8=Duck or goose fat

Description
of Data
variables in
nutritional
study





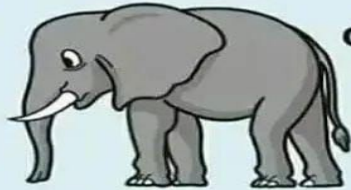
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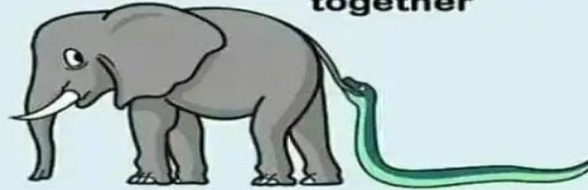
Statistics



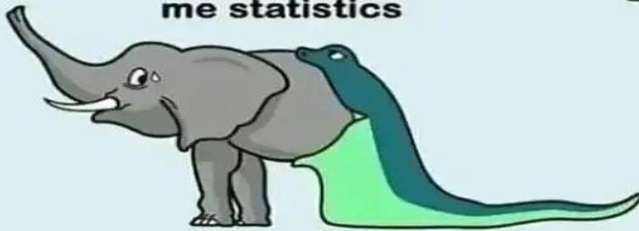
**Computer
Science**



**We will work
together**



**Please teach
me statistics**



**Now I m
DATA SCIENTIST**



Henceforth, Data science is a field that involves using statistical and computational techniques to extract insights and knowledge from data. Data Science is all about to Play with Data.